Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Plane Shapes: Their Types and Properties

A plane shape is a flat shape — one that lies on a single surface and has only length and width, no thickness. Every shape you draw on paper is a plane shape.

Plane shapes fall into two broad families by the lines that make their edges:

- **Polygons** are plane shapes whose sides are straight line segments. They are named by the number of sides they have.
- Shapes with **curved** edges, of which the most important is the **circle**.

A polygon is a **regular** polygon when all its sides are equal and all its angles are equal. When its sides or angles are not all equal, it is an **irregular** polygon.

Each named plane shape has its own set of properties — its sides, angles and corners. Learn these properties, because they are what let you tell one shape from another.

Triangle — a polygon with 3 sides and 3 angles. Its 3 angles always add up to 180°.

Quadrilaterals — polygons with 4 sides. Their 4 angles always add up to 360°.

- **Square** — 4 equal sides and 4 right angles (each 90°). Its 2 diagonals are equal and cross at right angles.
- **Rectangle** — 4 right angles, with opposite sides equal and parallel.
- **Parallelogram** — 2 pairs of opposite sides that are equal and parallel, but its angles are not necessarily right angles. Opposite angles are equal.
- **Trapezium** — a quadrilateral with only **one** pair of parallel sides.

Circle — a plane shape with a curved edge. Every point on its edge is the same distance from its centre. That distance is called the **radius**, and the distance across the circle through the centre is the **diameter**.

Seeing what shapes have **in common** (their similarities) and what makes each one **different** is just as important as naming them.

Similarities

- A square, rectangle, parallelogram and trapezium are all quadrilaterals: each has 4 sides and 4 angles, and each set of 4 angles adds to 360°.
- A square and a rectangle both have 4 right angles and 2 diagonals that are equal.

Differences

- A square has all 4 sides equal; a rectangle has only its opposite sides equal.
- A parallelogram can have no right angles at all, unlike a square or rectangle, which always have 4 right angles.
- A trapezium is the only quadrilateral that has just one pair of parallel sides; the square, rectangle and parallelogram each have two pairs.
- The circle is different from all of these because it has no sides and no corners — just one continuous curved edge.

Example — sorting a shape by its properties. Suppose you are told a plane shape has 4 sides, 4 right angles, and all 4 sides equal. Which shape is it?

Work through the clues one at a time.

1. "4 sides" — it is a quadrilateral, so it could be a square, rectangle, parallelogram or trapezium.
2. "4 right angles" — this rules out the parallelogram (whose angles need not be right angles) and the trapezium (which has no right angles at all). We are left with a square or a rectangle.
3. "All 4 sides equal" — a rectangle has only its opposite sides equal, so the only shape with 4 equal sides and 4 right angles is the **square**.

Answer: the shape is a square.

Practice

1. A plane shape has 4 sides, exactly one pair of parallel sides, and its 4 angles add up to 360°. Name the shape.
2. State one property that a square and a rectangle share, and one property in which they differ.
3. A plane shape has a curved edge, no straight sides and no corners. Which plane shape is it, and what do we call the distance from its centre to its edge?
4. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?

Section 2: Getting to Know the Circle and Its Parts

A circle is a plane shape in which every point on its edge is exactly the same distance from one fixed point inside it, called the centre. Because every point is the same distance from the centre, the circle has several named parts that describe lines, curves and regions around it. Learn each name with what it looks like, and you will be able to label any circle.

- The centre is the point inside the circle that all points on the circle are the same distance from.
- The radius (plural: radii) is a straight line from the centre to any point on the circle. Every radius has the same length.
- The diameter is a straight line that passes through the centre and touches the circle at two points. It is twice as long as the radius.
- The chord is a straight line joining two points on the circle. A chord does not have to pass through the centre — but if it does, it is a diameter.
- The circumference is the distance all the way around the circle, that is, its perimeter.
- An arc is part of the circumference, a curved piece of the circle's edge.
- A segment is the region of the circle cut off by a chord — the part between the chord and the arc above or below it.
- A sector is a region of the circle bounded by two radii and the arc between them, like a slice of a pie.
- A tangent is a straight line that touches the circle at exactly one point and does not pass through it.

Notice how the names pair up by what they are: the radius and diameter are straight lines from or through the centre; the chord and tangent are straight lines touching the circle; the circumference and arc belong to the circle's edge; and the segment and sector are regions inside the circle.

Let's label the parts of a circle together, in a useful order.

Draw a circle and mark its centre with a dot, O.

First the straight lines. Draw a line from O straight out to the circle — this is a radius. Now extend that line right across the circle, still through O, until it reaches the circle on the other side — this whole line is a diameter. It is twice as long as the radius. Now draw a different straight line joining two points on the circle but not passing through O — this is a chord.

Now the edge. The entire distance around the circle is the circumference. Take any curved part of that edge — that curved piece is an arc.

Now the regions. The slice between your diameter (or any two radii) and the arc between them is a sector. The part between a chord and the arc above it is a segment.

Finally, draw a straight line that just touches the circle at one point and does not go through it — this is a tangent.

So, in one picture, a circle can carry a radius, a diameter, a chord, a circumference, an arc, a segment, a sector and a tangent, each with a clear name.

Practice

1. A straight line joins two points on a circle but does not pass through the centre. What is this line called?

2. A slice of a circle is bounded by two radii and the arc between them. What is this region called?
3. A straight line touches a circle at exactly one point. What is this line called?
4. A circle has a radius of 4 cm. How long is its diameter?
5. A circle has a diameter of 10 cm. How long is its radius?
6. What name is given to the distance all the way around a circle?
7. A curved piece of a circle's edge is called what?
8. What is the region of a circle that is cut off by a chord called?

Section 3: Finding the Perimeter of Plane Shapes

The perimeter of a shape is the total distance around its boundary. For any straight-sided shape (a polygon), you find the perimeter by adding the lengths of all its sides. For a circle, the boundary is curved, and the distance around it is called the circumference — but it is still a perimeter, and you find it with a special formula.

Because some polygons have sides that are equal, we can often add them more quickly. Let us look at each shape you need to know.

Square and rectangle. A square has four equal sides. If each side is s, then its perimeter is
$perimeter = 4s.$
A rectangle has two lengths l and two breadths (widths) b, so its perimeter is
$perimeter = 2(l + b).$
For example, a rectangle of length 7 cm and breadth 4 cm has perimeter $2(7 + 4) = 2 \times 11 = 22$ cm.

Parallelogram and trapezium. A parallelogram has opposite sides equal: a base and a slant side. If the base is b and the slant side is a, then
$perimeter = 2(b + a).$
A trapezium has four sides, usually all different, so you simply add the four given sides together.

Regular polygons. A regular polygon has all sides equal. So if a regular polygon has n sides and each side is s, then
$perimeter = n \times s.$
A regular hexagon has 6 equal sides, so with sides of 5 cm its perimeter is $6 \times 5 = 30$ cm. A regular pentagon has 5 equal sides, so with sides of 5 cm its perimeter is $5 \times 5 = 25$ cm.

Circle. The distance around a circle is called the circumference. Its formula is
$circumference = 2\pi r,$
where r is the radius. Using $\pi = \frac{22}{7}$, a circle of radius 7 cm has circumference
$2 \times \frac{22}{7} \times 7 = 2 \times 22 = 44$ cm.$

Worked example 1 — square. Find the perimeter of a square of sides 8 cm.

A square has 4 equal sides, so
$perimeter = 4 \times 8 = 32$ cm.$

Worked example 2 — rectangle. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.

$$\text{perimeter} = 2(l + b) = 2(7 + 4) = 2 \times 11 = 22 \text{ cm}.$$

Worked example 3 — regular hexagon. Find the perimeter of a regular hexagon with sides 5 cm.

A regular hexagon has 6 equal sides, so
$perimeter = 6 \times 5 = 30$ cm.$

Worked example 4 — parallelogram. Find the perimeter of a parallelogram whose base is 6 cm and slant side is 5 cm.

Opposite sides of a parallelogram are equal, so the two bases are 6 cm and the two slant sides are 5 cm:
$perimeter = 2(6 + 5) = 2 \times 11 = 22$ cm.$

Worked example 5 — finding a missing side. The perimeter of a square is 36 cm. Find the length of one side.

A square has 4 equal sides, so one side is the perimeter divided by 4:
$s = \frac{36}{4} = 9$ cm.$

Worked example 6 — finding the radius from the circumference. The circumference of a circle is 22 cm. Find its radius, taking $\pi = \frac{22}{7}$.

Using circumference $= 2\pi r$:
$$22 = 2 \times \frac{22}{7} \times r.$$
Divide both sides by 2:
$$11 = \frac{22}{7} \times r.$$
So
$$r = 11 \times \frac{7}{22} = \frac{77}{22} = 3.5 \text{ cm}.$$

Worked example 7 — trapezium. A trapezium has sides of 6 cm, 4 cm, 8 cm and 5 cm. Find its perimeter.

Add all four sides:
$$6 + 4 + 8 + 5 = 23 \text{ cm}.$$

Worked example 8 — finding a missing length. A rectangle has a breadth of 3 cm and a perimeter of 22 cm. Find its length l.

\[\text{perimeter} = 2(l + b), \quad 22 = 2(l + 3).$$ Divide both sides by 2: $$11 = l + 3, \quad \text{so } l = 11 - 3 = 8 \text{ cm}.\]

Worked example 9 — a shape made of a rectangle and a semicircle. A figure is made of a rectangle 9 cm long and 5 cm high. Along the top of the rectangle sits a semicircle of radius 3.5 cm, with a 1 cm straight segment on each side of it. Find the perimeter of the whole figure, taking $\pi = \frac{22}{7}$.

Walk around the outside of the figure. You meet: the 9 cm bottom, the 5 cm left side, the 1 cm top-left segment, the curved semicircle, the 1 cm top-right segment, and the 5 cm right side.

The curved top is half a circle. Half of the circumference $2\pi r$ is πr, so
$$\text{semicircle} = \pi r = \frac{22}{7} \times 3.5 = 11 \text{ cm}.$$
Now add every outside part:
$$9 + 5 + 1 + 11 + 1 + 5 = 32 \text{ cm}.$$
The perimeter of the figure is 32 cm.

Practice

1. Find the perimeter of a square of sides 8 cm.
2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
3. What is the perimeter of a regular pentagon with sides 5 cm?
4. Find the perimeter of a parallelogram whose base is 6 cm and slant side is 5 cm.
5. The circumference of a circle is 22 cm. Find the radius of the circle, taking $\pi = 22/7$.
6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle (that is, its perimeter).
7. Find the perimeter of a regular hexagon with sides 5 cm.
8. A figure is made of a rectangle 9 cm long and 5 cm high. The top of the rectangle has a 1 cm straight segment on the left and a 1 cm straight segment on the right, and between them a semicircle (half a circle) of radius 3.5 cm opening upward. Find the perimeter of the whole figure, taking $\pi = 22/7$.

Check your work

Attempt every question before you read this page.

Plane Shapes: Their Types and Properties

1. A trapezium.

   "4 sides" tells us it is a quadrilateral, so its angles add to 360° — that clue does not single out one shape. The deciding property is "exactly one pair of parallel sides," which belongs only to the trapezium among the quadrilaterals.

2. Share: both have 4 right angles (and 4 sides). Differ: a square has all 4 sides equal, while a rectangle has only its opposite sides equal.
3. It is a circle, and the distance from the centre to the edge is called the radius.
4. 6 triangles.

   From one vertex of a polygon you can draw a diagonal to every vertex except the two next to it and itself. An octagon has 8 vertices, so from one vertex you draw diagonals to $8 - 3 = 5$ non-adjacent vertices, splitting it into $5 + 1 = 6$ triangles. In general, a polygon with n sides is split into $n - 2$ triangles, and $8 - 2 = 6$.

Getting to Know the Circle and Its Parts

1. A chord.
2. A sector.
3. A tangent.
4. 8 cm, because the diameter is twice the radius.
5. 5 cm, because the radius is half the diameter.
6. The distance all the way around the circle — its perimeter.
7. An arc (a curved part of the circumference).
8. The region between a chord and the arc above it.

Finding the Perimeter of Plane Shapes

1. 32 cm

   A square has 4 equal sides, so perimeter $= 4 \times 8 = 32$ cm.
2. 22 cm

   Perimeter $= 2(l + b) = 2(7 + 4) = 2 \times 11 = 22$ cm.
3. 25 cm

   A regular pentagon has 5 equal sides, so perimeter $= 5 \times 5 = 25$ cm.
4. 22 cm

   Perimeter $= 2(\text{base} + \text{slant side}) = 2(6 + 5) = 2 \times 11 = 22$ cm.
5. 3.5 cm

   Circumference $= 2\pi r$, so $22 = 2 \times (22/7) \times r$. Divide both sides by 2: $11 = (22/7) \times r$, so $r = 11 \times 7/22 = 77/22 = 3.5$ cm.
6. 6d

   Breadth $= d$ and length $= 2d$. Perimeter $= 2(\text{length} + \text{breadth}) = 2(2d + d) = 2 \times 3d = 6d$.
7. 30 cm

   A regular hexagon has 6 equal sides, so perimeter $= 6 \times 5 = 30$ cm.
8. 32 cm

   The curved top is a semicircle, whose length is half the circumference of a circle of radius 3.5 cm: $(1/2) \times 2\pi r = \pi r = (22/7) \times 3.5 = 11$ cm. Add the straight parts: bottom 9 cm, left side 5 cm, right side 5 cm, and the two 1 cm top segments. Perimeter $= 9 + 5 + 5 + 1 + 1 + 11 = 32$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-622615d1	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____